matrix eigenvalues (not fixed a priori):

$$\text{CortexCoords} \equiv \text{eigenvalues}(X),$$
$$X^\dagger = X, \qquad X \in M_{3\times3}(\mathbb{C}).$$

Proved in `BFSSIsomorphism.lean` *using Mathlib's* `Matrix.IsHermitian.eigenvalues`.

Connection to Schreiber's mHoTT: the modal operators of Modal Homotopy Type Theory ARE the Zoom Operators of the USF. The ∞-topos of mHoTT = the soma-field configuration space.

Type-safe zooming: a scale mismatch in Λ is a type error caught by the Lean 4 kernel at compile time — mathematically impossible to apply human-scale emotional operators to a galaxy.

III · Attractor Dynamics

Hopfield energy landscape:

$$H(\mathbf{e}) = -\tfrac{1}{2}\mathbf{e}^\top W \mathbf{e}$$

Emotional states are attractor basins; trauma wells are deep narrow minima; therapeutic change is landscape modification.

FM-HN Correspondence Principle (*Lean 4 kernel-verified*):

$$\beta(t) = \beta_0 + \kappa\,|\Phi(t)|^2 \xrightarrow[\Phi\to 0]{} \beta_0$$

Under zero somatic stress, FM-HN reduces exactly to the classical 1982 Hopfield network. The 1982 and 2020 Hopfield networks are both limiting cases.

Consciousness threshold T_c: the spectral gap of the somatic field operator opens at $|\phi| > T_c$. Below: diffusive dynamics, no stable attractors, no experience. Above: ordered phase, stable phenomenal states (qualia = attractor basin topology).

Kramers mean first-passage time:

$$\langle T\rangle = \frac{2\pi}{\omega_A\,\omega_B}\,e^{\Delta V/D}$$

Trauma asymmetry: formation time $\ll$ dissolution time. Formation: barrier crossed from above (external forcing). Escape: from inside the well, exponentially slow for $\Delta V \gg D$.

WKB tunnelling amplitude:

$$P \approx \exp\left(-\frac{2}{\hbar_{\text{geo}}}\int_{q_1}^{q_2}\sqrt{V(q)-E}\,dq\right)$$

Quantum annealing experiment (QUANT-EXP-1): D-Wave reaches Awe basin in 3/3 barrier cases; classical simulation achieves it in 0/48. Quantum advantage is real.

Arnold tongue — rapport as Huygens frequency locking. Width = social/attentional bandwidth:

$$|\omega_{\text{ext}} - \omega_0| < \Delta\omega_{\text{lock}}(\kappa)$$

SHO frequency (*physlib* `trajectory_equationOfMotion`):

$$\omega^2 = k/m, \quad \omega^2 = v^2k^2 \quad \text{(dispersion relation)}$$

IV · The 11D Decomposition — M-Theory / BFSS / Hořava-Witten

$$M_{11} = M_4 \times X_7, \quad X_7 = D_{5\text{–}7}\times D_8 \times D_{9\text{–}11}$$

- $D_{1\text{–}4}$: **Spacetime** — body embedded in Lorentzian 3+1D; symmetry group $SO(3,1)$
- $D_{5\text{–}7}$: **Propagator** — somatic EM field; gauge field trapped on D3-brane worldvolume
- D_8: **Limbic axis** — Hořava-Witten orbifold $S^1/\mathbb{Z}_2$, segment $[-1,+1]$; body at -1, mind at $+1$; trauma = topological obstruction
- $D_{9\text{–}11}$: **Cortex** — BFSS matrix eigenvalue spectrum; emergent geometry, not fixed coordinates

BFSS identification (Banks-Fischler-Shenker-Susskind 1997): cortex coordinates are NOT fundamental — they emerge from the eigenvalues of $N \times N$ Hermitian matrices. Thoughts are eigenvalues; social dynamics are matrix commutators.

Hořava-Witten (1996): D_8 as the compact orbifold separates two 10D boundary spacetimes. The limbic system is the physical thickness between biology and psychology.

Cosmological limit: $\kappa_{\text{bio}} \to 0$ recovers the linearised Einstein equation. The cosmological constant $\Lambda \equiv \langle \text{tr}\,\Phi\rangle_0$ — the vacuum expectation of the somatic tensor trace.

SHO derivation: $\omega^2 = k/m$ is derived (not postulated) as the lowest-order term in the Taylor expansion of the Calabi-Yau moduli space metric. This constrains the compactification geometry.

Organism hierarchy:

$$11D \xrightarrow{\text{project}} 7D \xrightarrow{\text{project}} 4D$$

A rock is a 4D organism. A jellyfish is a 7D organism. A human is an 11D organism.